

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

Kopiereg voorbehou

Semester test 1

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Module EBN 122

Elektrisiteit en Elektronika

19 Augustus 2011

Module EBN 122

Electricity and Electronics

19 August 2011



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
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Memo 1

Toetsinligting: Test information:			
Maksimum punte: <i>Maximum marks</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute 90 minutes	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		16	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.

Dosent(e): <i>Lecturer(s):</i>	1 Prof. J. Joubert
	2 Mr. W. Badenhorst
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>	
Groephoof: <i>Group Head:</i>	Prof. J. Joubert
	Handtekening: <i>Signature:</i>

VRAAG 1 / QUESTION 1 [14]

Vraag 1.1 [1] 'n 24 V battery wat 30 A kan lewer word gebruik as energiebron om een liter water te kook. Gestel dit neem ongeveer 100 Wh om 1 liter kraanwater tot by kookpunt te kry. Hoeveel minute sal dit neem vir die water om te kook?	Question 1.1 [1] A 24 V battery that can deliver 30 A is used as energy source to boil one liter of water. Assuming it requires 100 Wh to bring 1 liter of tap water to boiling point, how many minutes will it take for the water to boil?
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$$W = VI \cdot t$$

$$100 \text{ Wh} \cdot \frac{60 \text{ min}}{1 \text{ h}} = 24 \cdot 30 \cdot t$$

$$t = 8 \text{ min } 20 \text{ sec}$$

$$= \underline{8.3 \text{ min}} \checkmark$$

Vraag 1.2 [1] As deel van 'n energiebesparingsprogram is daar in 'n winkelsentrum besluit om alle ou 100 W gloeilampe met 14 W CFL gloeilampe te vervang. Daar is 50 gloeilampe vervang wat elk gemiddeld 7 ure per dag brand. Indien die elektrisiteitsprys 80c/kWh is, hoeveel geld bespaar die sentrum per 30-dag maand?	Question 1.2 [1] As part of an energy savings program a commercial centre decided to replace all the old 100 W globes with 14 W CFLs. They replaced 50 globes each being on for 7 hours per day on average. If the price of electricity is 80c/kWh, how much money does the centre save per 30-day month?
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$$\text{Power saved} = 100 - 14 = 86 \text{ W}$$

$$\text{Energy} = 86 \cdot 50 \cdot 7 \text{ h} \cdot 30 \text{ days}$$

$$= 903 \text{ kWh}$$

$$\text{Cost} = 903 \text{ kWh} \cdot 0.8 \text{ R/kWh}$$

$$= \underline{722.4 \text{ p.m}} \checkmark$$

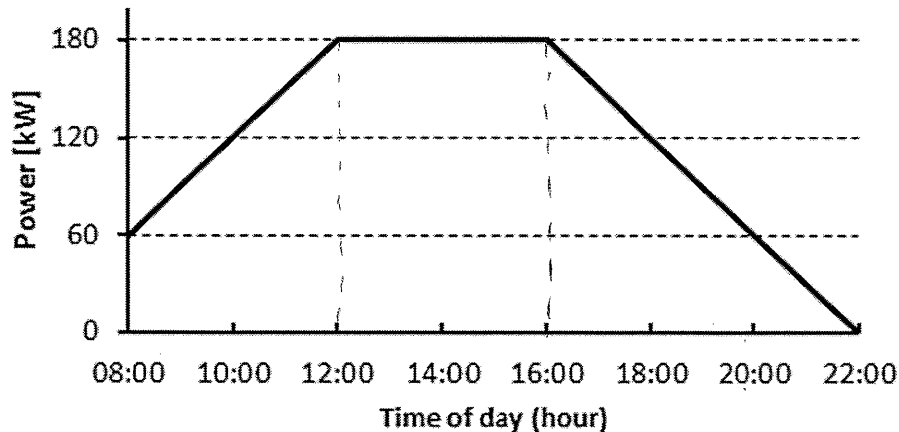
Vraag 1.3 [1]

Die gemiddelde drywingslasprofiel van 'n besigheid tydens besigheidsure word in die grafiek getoon. Bereken hoeveel energie die besigheid per dag verbruik.

Question 1.3 [1]

The average power load profile of a business during business hours is shown in the graph. Calculate the amount of energy the business consumes per day.

Lasprofiel / Load Profile



$$W = \int P dt = 4 \cdot 60 + 4 \cdot 120 \cdot \frac{1}{2} + 4 \cdot 180 + 6 \cdot 180 \cdot \frac{1}{2} = \underline{1740 \text{ kWh}}$$

Vraag 1.4 [1]

Die lading in 'n kapasitor word gemeet as 200 mC. 10 minute later word die lading gemeet as 140 mC. Wat is die gemiddelde lekstroom deur die kapasitor?

Question 1.4 [1]

The charge in a capacitor is measured at 200 mC. 10 minutes later the charge is measured at 140 mC. What is the average leakage current through the capacitor?

$$I = \frac{\Delta Q}{\Delta t} = \frac{(200 - 140) \cdot 10^{-3}}{10 \cdot 60} = \frac{60 \cdot 10^{-3}}{600} = \underline{100 \mu A}$$

Vraag 1.5 [6]

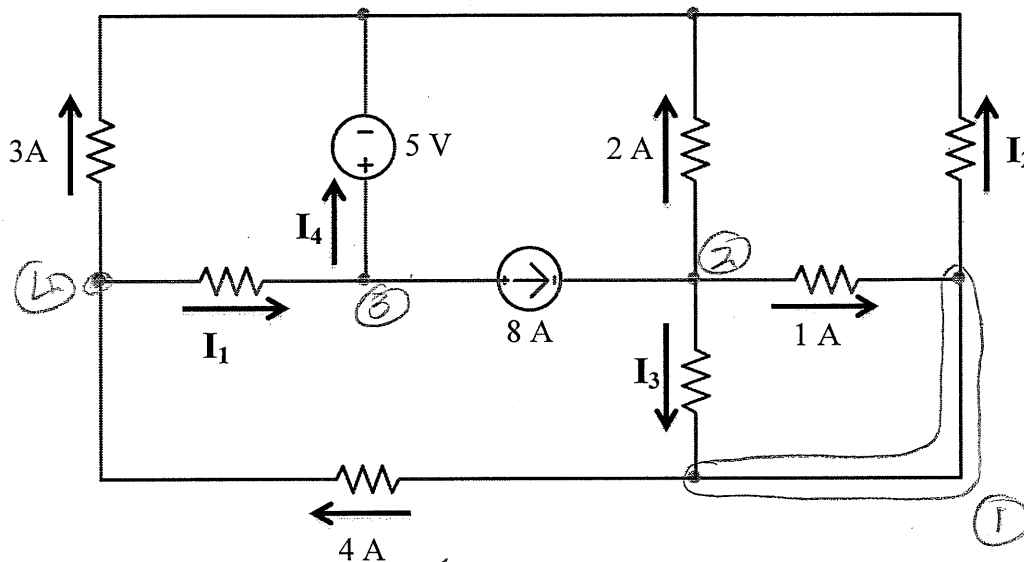
Bepaal:

- a) Die aantal Takke. (1)
- b) Aantal Nodes. (1)
- c) I_2 (2)
- d) Die drywing geassosieer met die 5V bron. (2)

Question 1.5 [6]

Determine:

- a) The number of Branches. (1)
- b) The number of Nodes. (1)
- c) I_2 (2)
- d) The power associated with the 5V source. (2)



a) Branches = 9 ✓

b) Nodes = 5 ✓

c) At node ①: $I_3 + 1 = 4 + I_2$
 $I_2 = I_3 - 3$

at node ②: $8 = 2 + 1 + I_3$
 $I_3 = 5$

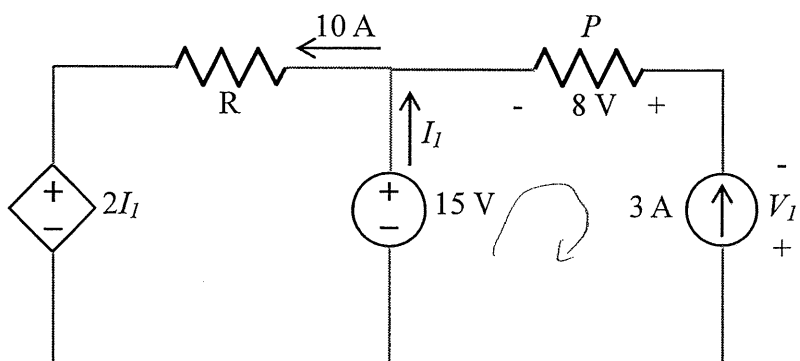
$\therefore I_2 = 2A$ ✓✓

d) ③ $I_1 = I_4 + 8$

@ ④ $4 = I_1 + 3 \therefore I_1 = 1A$

$\therefore I_4 = 1 - 8 = -7A$

$\therefore P = VI = 5 \cdot (-7) = -35W$ ✓✓

Vraag 1.6 [4]Bereken V_I , I_I , P en R .**Question 1.6 [4]**Calculate V_I , I_I , P and R .

$$\text{KVL: } -V_I - 15 - 8 = 0$$

$$V_I = \underline{-23 \text{ V}}, \checkmark$$

$$\text{KCL: } I_I + 3 = 10$$

$$I_I = \underline{7 \text{ A}}, \checkmark$$

$$P = 8 \cdot 3 = \underline{24 \text{ W}}, \checkmark$$

$$R = \frac{V}{I} = \frac{15 - 2 \cdot 7}{10} = \underline{0.1 \Omega}, \checkmark$$

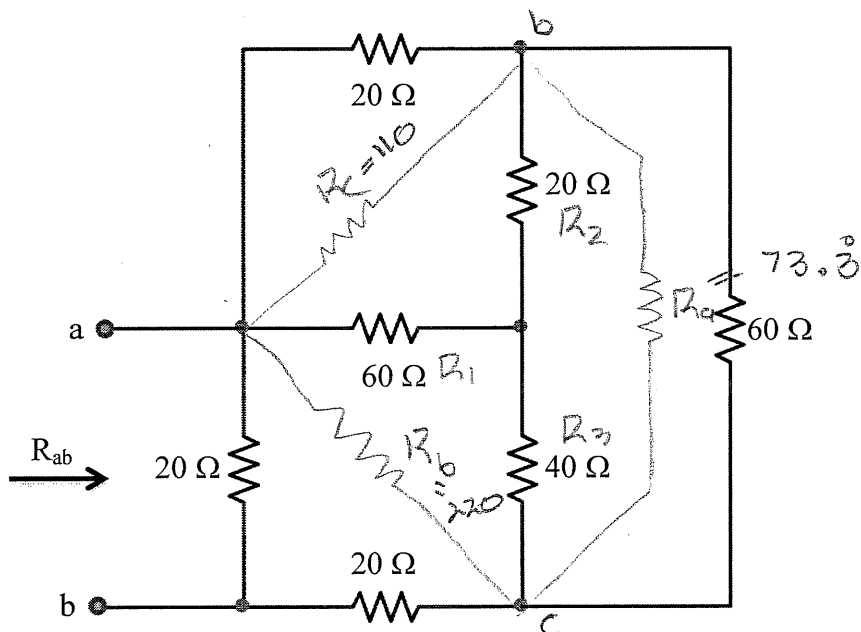
VRAAG 2 / QUESTION 2 [14]

Vraag 2.1 [3]

Bereken die ekwivalente weerstand R_{ab} .

Question 2.1 [3]

Calculate the equivalent resistance R_{ab}


$$R_A = 3R_f$$

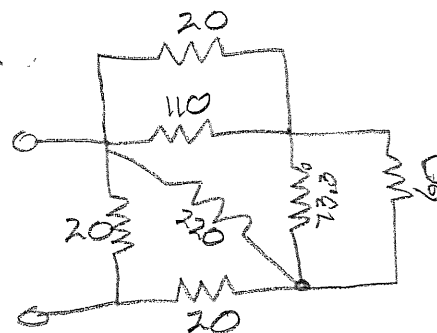
Note can do $\Delta \rightarrow \gamma$ or $\gamma \rightarrow \Delta$

For $\gamma \rightarrow \Delta$

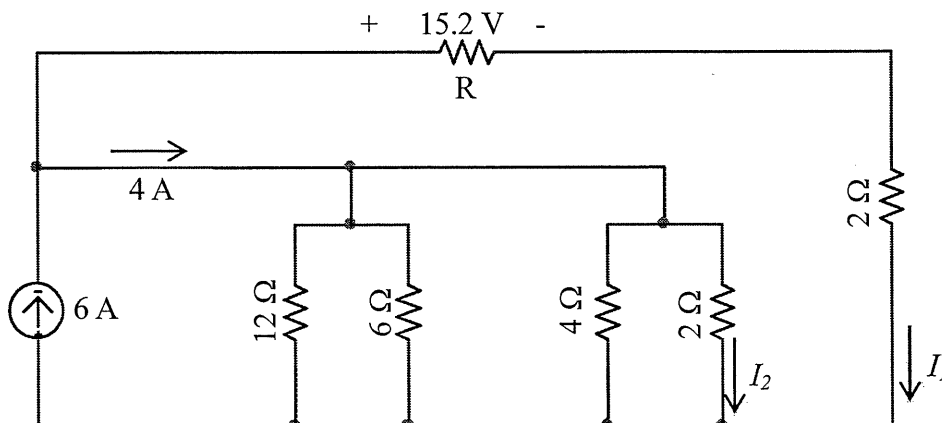
$$\begin{aligned} R_a &= \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1} \\ &= \frac{60 \cdot 20 + 20 \cdot 40 + 60 \cdot 40}{60} \\ &= \frac{4400}{60} \\ &= 73.3 \Omega \end{aligned}$$

$$R_b = \frac{4400}{20} = 220 \Omega$$

$$R_c = \frac{4400}{40} = \underline{110 \Omega}$$



$$\begin{aligned} & 20 // 110 + 60 // 73.3 \\ & = 49.923 \\ & (220 // 49.923 + 20) // 20 \\ & = 15.04 \Omega \end{aligned}$$

Vraag 2.2 [3]Bereken: I_1 , I_2 en R .**Question 2.2 [3]**Calculate: I_1 , I_2 and R .KCL:

$$I_1 = 6 - 4 = 2A \quad \checkmark$$

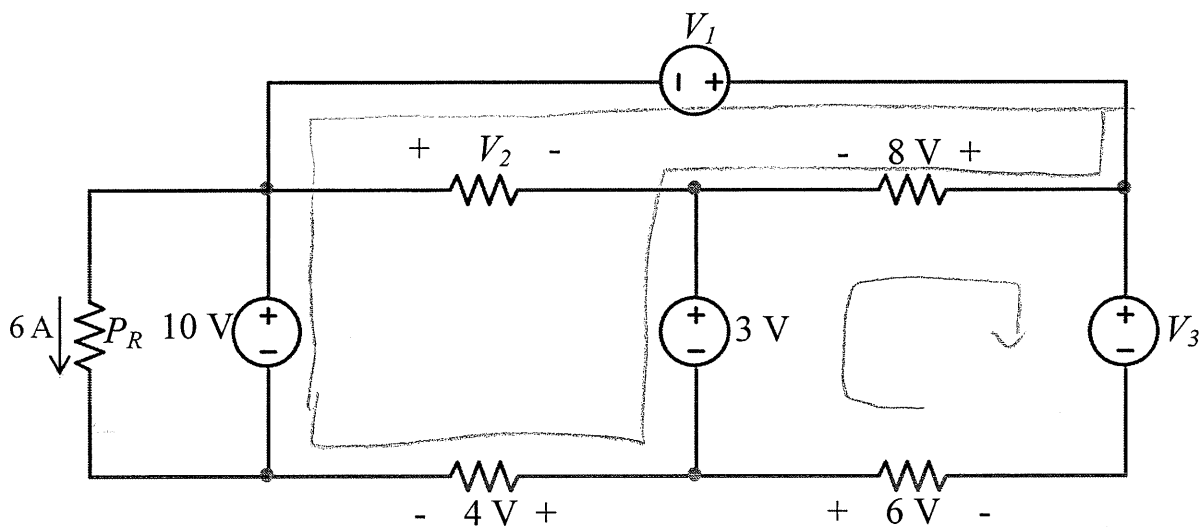
$$\therefore R = \frac{V}{I} = \frac{15.2}{2} = 7.6\Omega \quad \checkmark$$

$$I_2 = 4 \cdot \frac{\frac{1}{\frac{1}{12} + \frac{1}{6} + \frac{1}{4} + \frac{1}{2}}} = 4 \cdot \frac{1}{1} = 2A \quad \checkmark$$

or

$$\begin{aligned} I_2 &= 4 \cdot \frac{R_{11}}{R_{11} + 2} \\ &= 4 \cdot \frac{2}{2+2} \\ &= 2A \quad \checkmark \end{aligned}$$

$$\begin{aligned} R_{11} &= 12 \parallel 6 \parallel 4 \\ &= 2\Omega \end{aligned}$$

Vraag 2.3 [3]Bereken: V_1 , V_3 en die drywing P_R .**Question 2.3 [3]**Calculate: V_1 , V_3 and the power P_R .

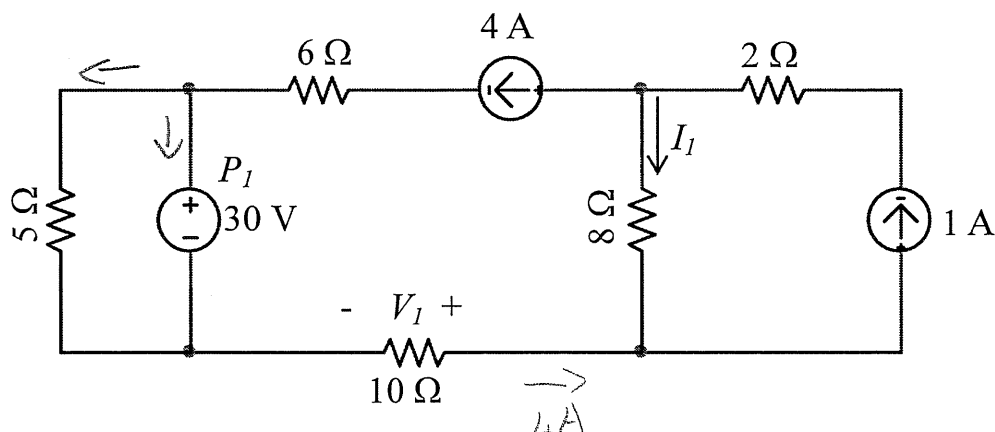
$$\text{KVL 1: } -10 - V_1 + 8 + 3 + 4 = 0$$

$$\underline{V_1 = 5V} \checkmark$$

$$\text{KVL 2: } V_3 - 6 - 3 - 8 = 0$$

$$\underline{V_3 = 17V} \checkmark$$

$$P_R = VI = 10 \cdot 6 = \underline{60W} \checkmark$$

Vraag 2.4 [3]Bereken: V_I , I_I en P_I t.o.v. die 30 V bron.**Question 2.4 [3]**Calculate: V_I , I_I and P_I w.r.t. the 30 V source.

$$I_I + 4 = 1 \text{ A}$$

$$I_I = -3 \text{ A}, \quad \checkmark$$

$$V_I = -4 \cdot 10 = -40 \text{ V}, \quad \checkmark$$

P_I :

$$\text{KCL: } \frac{30}{5} + I = 4$$

$$I = 4 - 6$$

$$= -2 \text{ A}$$

$$P_I = VI$$

$$= 30 \cdot -2$$

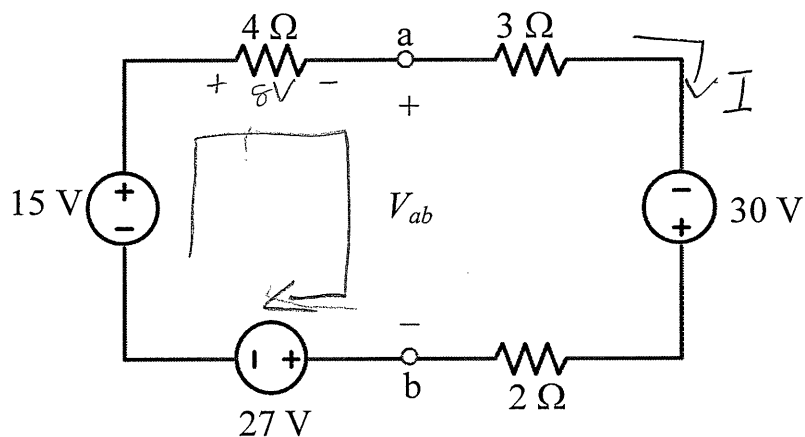
$$= -60 \text{ W}, \quad \checkmark$$

Vraag 2.5 [2]

Bepaal V_{ab} .

Question 2.5 [2]

Determine V_{ab} .



$$I = \frac{V}{R} = \frac{30 + 15 - 27}{4 + 3 + 2} = \frac{18}{9} = 2A$$

$$\text{KVL: } -15 + 4I + V_{ab} + 27 = 0$$

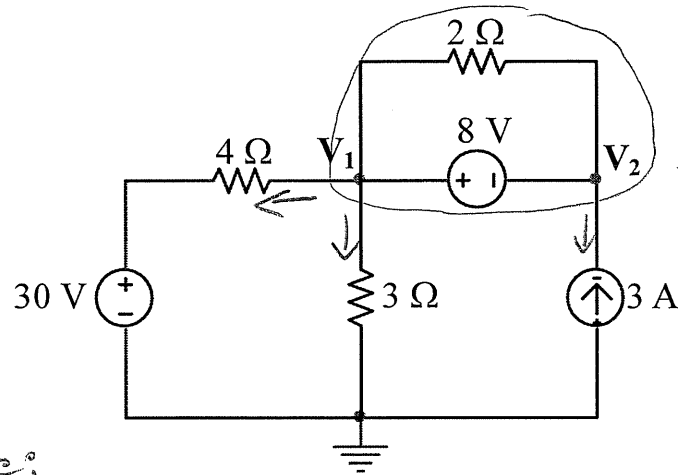
$$\begin{aligned} V_{ab} &= 15 - 4 \cdot 2 - 27 \\ &= \underline{-20V} \quad \checkmark \checkmark \end{aligned}$$

VRAAG 3 / QUESTION 3 [17]

Vraag 3.1 [2]

Gebruik node-analise en bereken V_1 en V_2 .

Question 3.1 [2]



Supernode:
 $V_1 - V_2 = 8V \quad \textcircled{1}$

KCL @ Supernode:
 $\frac{V_1 - 30}{4} + \frac{V_1}{3} + (-3) = 0$

$\times 12: 3V_1 - 90 + 4V_1 - 36 = 0$
 $7V_1 = 126$

$V_1 = \underline{18V} \rightarrow \checkmark$

$V_2 = V_1 - 8 = \underline{10V} \rightarrow \checkmark$

Vraag 3.2 [4]

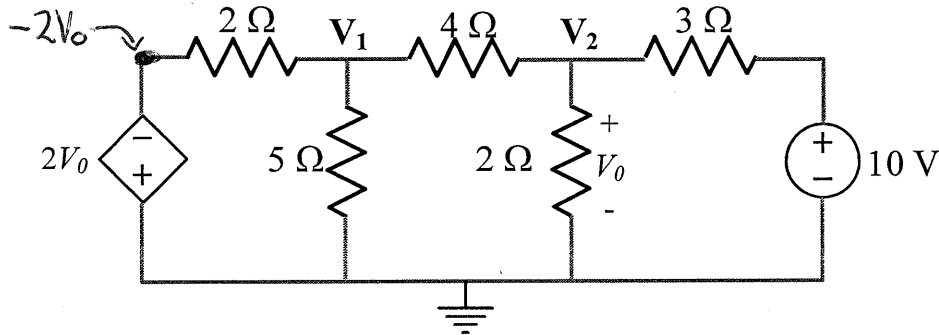
Gebruik node-analise om twee vergelykings in die volgende vorm op te stel:

$$\begin{aligned} aV_1 + bV_2 &= 0 \\ cV_1 + dV_2 &= 40 \end{aligned}$$

Question 3.2 [4]

Use node analysis to formulate two equations of the following form:

$$\begin{aligned} aV_1 + bV_2 &= 0 \\ cV_1 + dV_2 &= 40 \end{aligned}$$



$$\textcircled{1} \quad V_1 - \frac{(-2V_0)}{2} + \frac{V_1 - V_2}{4} + \frac{V_1}{5} = 0 \quad \text{but } V_0 = V_2$$

$$\begin{aligned} \times 20 \quad 10V_1 + 20V_2 + 5V_1 - 5V_2 + 4V_1 &= 0 \\ 19V_1 + 15V_2 &= 0 \quad \textcircled{1} \end{aligned}$$

$$\textcircled{2} \quad \frac{V_2 - V_1}{4} + \frac{V_2 - 10}{3} + \frac{V_2}{2} = 0$$

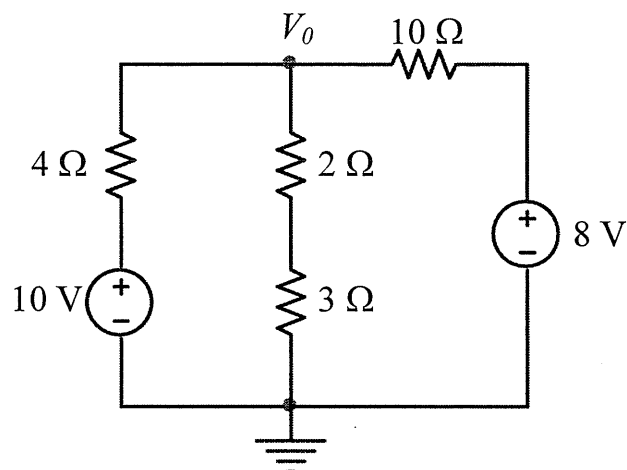
$$\begin{aligned} \times 12 \quad 3V_2 - 3V_1 + 4V_2 - 40 + 6V_2 &= 0 \\ -3V_1 + 13V_2 &= 40 \end{aligned}$$

$$a = 19 \checkmark \quad \text{or } a = -19 \text{ and } b = -15$$

$$b = 15 \checkmark$$

$$c = -3 \checkmark$$

$$d = 13 \checkmark$$

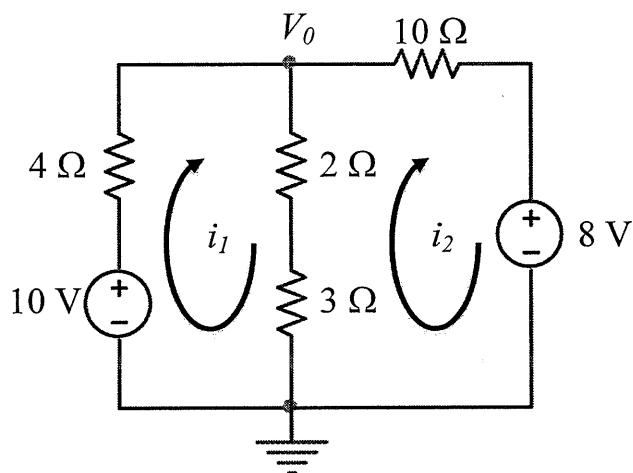
Vraag 3.3 [2]Gebruik **node**-analyse en bereken V_0 .**Question 3.3 [2]**Use **node** analysis and calculate V_0 .

$$\frac{V_0 - 10}{4} + \frac{V_0}{5} + \frac{V_0 - 8}{10} = 0$$

$$\times 20 \quad 5V_0 - 50 + 4V_0 + 2V_0 - 16 = 0$$

$$11V_0 = 66$$

$$V_0 = 6V \quad \checkmark \checkmark$$

Vraag 3.4 [2]Gebruik **lus**-analyse en bereken i_1 and V_0 .**Question 3.4 [2]**Use **mesh**-analysis and calculate i_1 and V_0 .Lus 1

$$-10 + 4\bar{i}_1 + 5(\bar{i}_1 - \bar{i}_2) = 0$$

$$9\bar{i}_1 - 5\bar{i}_2 = 10 \quad (1)$$

Lus 2

$$5(\bar{i}_2 - \bar{i}_1) + 10\bar{i}_2 + 8 = 0$$

$$-5\bar{i}_1 + 15\bar{i}_2 = -8 \quad (2)$$

$$(2) + 3(1) : (-5 + 27)\bar{i}_1 + (15 - 15)\bar{i}_2 = -8 + 30$$

$$22\bar{i}_1 = 22$$

$$\bar{i}_1 = \underline{1A} \quad \checkmark \checkmark$$

$$\text{KVL: } -10 + 4\bar{i}_1 + V_0 = 0$$

$$V_0 = 10 - 4$$

$$= \underline{6V}$$

Vraag 3.5 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$12i_1 + ai_2 = b$$

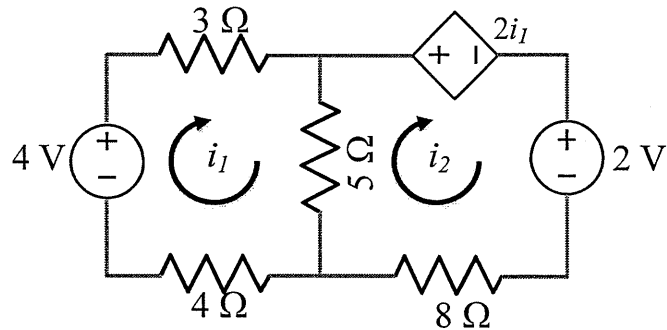
$$ci_1 + di_2 = -2$$

Question 3.5 [4]

Use mesh-analysis to formulate two equations of the following form:

$$12i_1 + ai_2 = b$$

$$ci_1 + di_2 = -2$$



Lus 1

$$-4 + 3i_1 + 5(i_1 - i_2) + 4i_1 = 0$$

$$12i_1 - 5i_2 = 4 \quad (1)$$

Lus 2

$$5(i_2 - i_1) + 2i_1 + 2 + 8i_2 = 0$$

$$-3i_1 + 13i_2 = -2 \quad (2)$$

$$a = -5 \checkmark$$

$$b = 4 \checkmark$$

$$c = -3 \checkmark$$

$$d = 13 \checkmark$$

Vraag 3.6 [3]

Los die volgende vergelyking op dmv Cramer se reël deur die volgende te bereken:

Δ , Δ_3 en V_3 .

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 1 \\ 2 & 0 & 5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$$

Question 3.6 [3]

Solve the following equation using Cramer's rule by calculating:

Δ , Δ_3 en V_3 .

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 1 \\ 2 & 0 & 5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 3 & 0 & 1 \\ 0 & 4 & 1 \\ 2 & 0 & 5 \end{vmatrix} = 3 \cdot 4 \cdot 5 + 0 + 0 - 1 \cdot 4 \cdot 2 - 0 - 0$$

$$= 60 - 8$$

$$= 52 \checkmark$$

$$\Delta_3 = \begin{vmatrix} 3 & 0 & 5 \\ 0 & 4 & 2 \\ 2 & 0 & 0 \end{vmatrix} = 0 + 0 + 0 - 5 \cdot 4 \cdot 2 - 0 - 0$$

$$= -40 \checkmark$$

$$\therefore V_3 = \frac{\Delta_3}{\Delta} = \frac{-40}{52} = -0.77 \checkmark$$